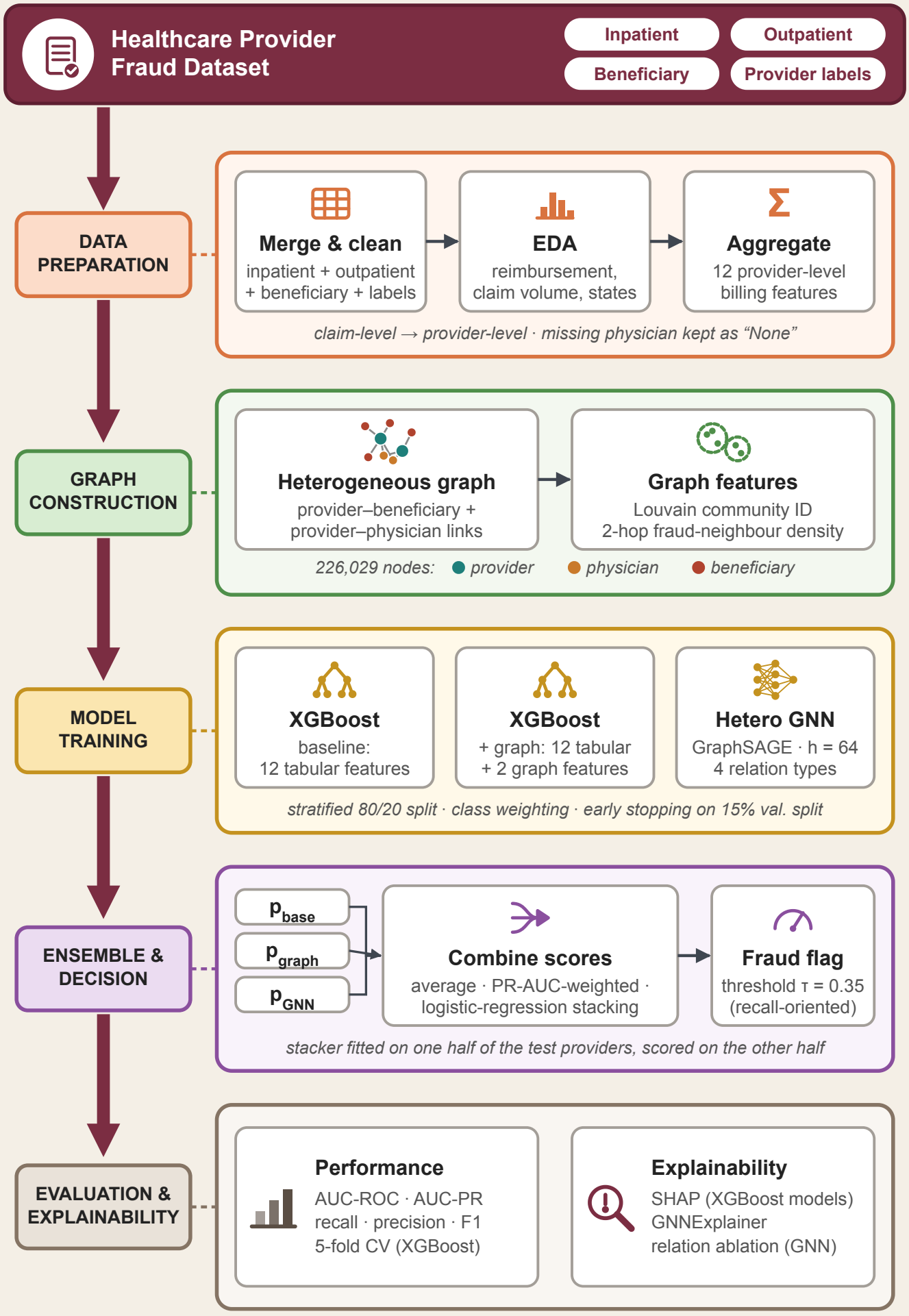

Healthcare Provider Fraud Dataset
Inpatient
Outpatient
Beneficiary
Provider labels
DATA PREPARATION
Merge & clean
inpatient + outpatient + beneficiary + labels
EDA
reimbursement, claim volume, states
Aggregate
12 provider-level billing features
claim-level → provider-level · missing physician kept as "None"
GRAPH CONSTRUCTION
Heterogeneous graph
provider–beneficiary + provider–physician links
Graph features
Louvain community ID
2-hop fraud-neighbour density
226,029 nodes: provider physician beneficiary
MODEL TRAINING
XGBoost
baseline: 12 tabular features
XGBoost
+ graph: 12 tabular + 2 graph features
Hetero GNN
GraphSAGE · h = 64
4 relation types
stratified 80/20 split · class weighting · early stopping on 15% val. split
ENSEMBLE & DECISION
p base
p graph
p GNN
Combine scores
average · PR-AUC-weighted · logistic-regression stacking
Fraud flag
threshold τ = 0.35 (recall-oriented)
stacker fitted on one half of the test providers, scored on the other half
EVALUATION & EXPLAINABILITY
Performance
AUC-ROC · AUC-PR
recall · precision · F1
5-fold CV (XGBoost)
Explainability
SHAP (XGBoost models)
GNNExplainer
relation ablation (GNN)